

Peripheral T-cell lymphoma

Nodal PTCL · ALCL · NK/T · ATLL · rare extranodal subtypes — **CD30 and ALK status pick the induction; ALK+ ALCL is the only subtype CHOP-like therapy reliably cures**

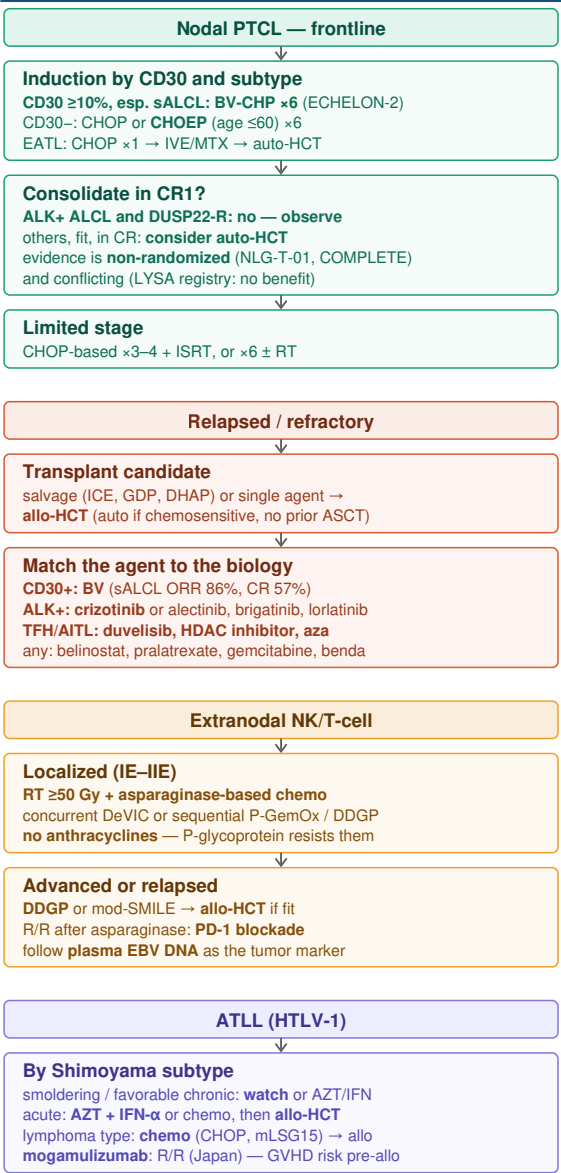
Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin, T-cell lymphoma · NCCN · primary trial reports and FDA actions verified against PubMed, September 2026.

ENTITIES — WHO-HAEM5 / ICC 2022

Entity	What defines it
PTCL-NOS ~26%	Diagnosis of exclusion; CD4 > CD8, loss of CD5/CD7 common. GATA3-high subset does worse than TBX21
Nodal TFH lymphoma, angioimmunoblastic type ~19%	Formerly AITL (ICC: follicular helper T-cell lymphoma). Older adults; rash, effusions, polyclonal hypergammaglobulinemia, Coombs+ hemolysis , EBV+ B-blasts. PD-1, CXCL13, ICOS, BCL6, CD10, TET2, DNMT3A, IDH2 R172, RHOA G17V . Follicular and NOS types share the TFH phenotype
ALCL, ALK+	Younger; NPM1::ALK t(2;5); strong uniform CD30. 5-yr OS ~85% — the good one
ALCL, ALK–	Test FISH: DUSP22-R (~30%) 5-yr OS 90% · TP63-R (~8%) 17% · neither 42%
Breast implant-associated ALCL	Textured implants ; late seroma or capsular mass, CD30+ ALK–. Tap the seroma → cytology + CD30
Extranodal NK/T-cell ~10%	WHO-HAEM5 drops “nasal type.” EBV+ always , CD56+, cCD3ε+, surface CD3–; angiocentric, necrotic mid-face mass; Asia and Latin America
ATLL ~10%	HTLV-1 ; Japan, Caribbean. CD4+CD25+ “flower cells,” hypercalcemia , lytic lesions, skin
EATL · MEITL	EATL: celiac (refractory sprue). MEITL (old “type II EATL”): no celiac, CD8+CD56+, SETD2 loss. Perforation risk
Hepatosplenic	γδ TCR; young men, IBD on anti-TNF + thiopurine ; hepatosplenomegaly, sinusoidal marrow, no nodes. Dismal

Frequencies from the International T-cell Project (Vose, JCO 2008, n=1,153) — **10% of referral diagnoses were misclassified**, so insist on expert hematopathology review.

TREATMENT PATHWAY



WORKUP

- **Excisional or core biopsy** with T-cell panel, **CD30** (% of cells), ALK, TFH markers, EBER; **TCR clonality**. ALK– ALCL → **DUSP22 and TP63 FISH**
- **PET/CT**, marrow biopsy, LDH, **HTLV-1 serology** (and calcium) in anyone at risk; **plasma EBV DNA** in NK/T and AITL
- HBV/HCV/HIV; echo before anthracycline. **Nasal NK/T: MRI of the nasopharynx** + ENT exam
- **ATLL: screen for Strongyloides** (co-endemic with HTLV-1) before steroids

PROGNOSIS — PIT AND IPI

PIT factors	5-yr OS	10-yr OS
0	62%	55%
1	53%	39%
2	33%	18%
3–4	18%	13%

PIT (Gallamini, Blood 2004): **age >60, LDH >ULN, ECOG ≥2, marrow involvement**. The **IPI** also stratifies PTCL and is the stratifier in ECHELON-2. **NK/T uses PINK** (age >60, stage III–IV, distant nodes, non-nasal primary; 3-yr OS 81/62/25% for 0/1/≥2); **PINK-E** adds detectable EBV DNA.

ATLL AND RARE SUBTYPES

- **ATLL subtypes (Shimoyama)**: acute, lymphoma, chronic (unfavorable if LDH, BUN or albumin abnormal), smoldering. **AZT + IFN-α** (meta-analysis, n=254): 5-yr OS 46% with first-line antivirals — benefit in **acute, chronic and smoldering**, while **lymphoma type does better with chemo**
- **mLSG15** (VCAP-AMP-VECP) beat biweekly CHOP on CR (40% vs 25%), 3-yr OS 24% vs 13% (NS). **Mogamulizumab**: R/R ATLL ORR 50% (Japan-approved, not US). **Given <50 d before allo-HCT it raises severe GVHD and NRM** (1-yr OS 32% vs 49%)
- **BIA-ALCL**: disease confined to capsule/seroma → **total en bloc capsulectomy + implant removal** (both sides) is usually curative; nodal or unresectable → BV-CHP
- **Hepatosplenic**: CHOP fails — ICE or IVAC-type induction, then **allo-HCT in CR1**
- **EATL/MEITL**: nutrition + surgical review for perforation; IVE/MTX (Newcastle) → auto-HCT. **AITL**: steroids/cyclosporine for immune flares; watch for concurrent EBV+ B-cell lymphoma

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Page 2 — frontline evidence, relapsed disease, NK/T-cell lymphoma and toxicity

FRONTLINE EVIDENCE — WHAT IT ACTUALLY SHOWS

- **ECHELON-2 (BV-CHP vs CHOP, CD30 ≥10%)**: PFS HR 0.71, **OS HR 0.66** — the only regimen to beat CHOP on survival. 5-yr PFS **51% vs 43%**, OS **70% vs 61%**
- **But the benefit sits in systemic ALCL** (70% of enrollment; 5-yr PFS HR 0.55). PTCL-NOS HR 0.75 (NS); **AITL favored CHOP** (small n). The label is **sALCL or other CD30-expressing PTCL**; outside ALCL it is a reasonable, not proven, choice
- **CD30 threshold was ≥10%** by local IHC; retreatment with BV at relapse after BV-CHP still gave ORR 59%
- **CHOEP** (DSHNHL pooled analysis): etoposide improved 3-yr EFS **75% vs 51%** in age ≤60 with normal LDH; **over 60, CHOP-21 stays standard**
- **Adding to CHOP has failed**: Ro-CHOP (PFS 12.0 vs 10.2 mo, HR 0.81, **NS**, more toxicity); alemtuzumab-CHOP; lenalidomide-CHOP in AITL
- **Auto-HCT in CR1 — never randomized**. NLG-T-01 (CHOEP-14 → ASCT, ALK+ excluded): 5-yr OS 51%, PFS 44%, TRM 4%. COMPLETE cohort: ASCT in CR1 OS HR 0.37, driven by AITL. LYSA propensity-matched registry: **no PFS or OS benefit** (HR 1.02). Offer it to fit non-ALK+ patients in CR, especially AITL and higher IPI

EXTRANODAL NK/T-CELL LYMPHOMA

- **Anthracyclines fail** — tumor cells express P-glycoprotein (MDR1). Build regimens on **L-asparaginase/pegaspargase**, methotrexate, gemcitabine, platinum, etoposide
- **Localized**: radiation is the backbone. **DeVIC x3 + concurrent RT 50 Gy** (JCOG0211): 2-yr OS **78%** vs 45% historical RT alone. Sequential/sandwich P-GemOx or DDGP + RT is the common alternative
- **Advanced**: **DDGP beat SMILE** in a randomized trial (n=87; PFS HR 0.42, OS HR 0.41, 5-yr OS 74% vs 52%, less myelosuppression). SMILE: ORR 79% but **grade 4 neutropenia 92%** — require ALC ≥500
- **EBV DNA** in plasma: detectable at diagnosis is adverse (PINK-E); **persistent after therapy predicts relapse**
- **R/R after asparaginase**: **PD-1 blockade** — pembrolizumab responded in 7/7 (Kwong, Blood 2017); off-label in the US. Consolidate responders with allo-HCT

RELAPSED / REFRACTORY AGENTS — AND THEIR US STATUS

Agent	Evidence and status (Sept 2026)
Brentuximab vedotin <i>anti-CD30 ADC</i>	R/R sALCL ORR 86% , CR 57% (Pro, JCO 2012). Regular approval. Activity in CD30+ non-ALCL is lower and not CD30-level dependent
Pralatrexate <i>antifolate</i>	PROPEL: ORR 29% , DOR 10.1 mo, PFS 3.5 mo. Accelerated approval 2009 — still unconfirmed ; the confirmatory trial will not read out before ~2030
Belinostat <i>pan-HDAC</i>	BELIEF: ORR 26% , CR 11%, DOR 13.6 mo, but PFS 1.6 mo. Accelerated 2014, unconfirmed — ODAC (Nov 2023) criticized both delays. Least myelosuppressive; OK with low platelets
Romidepsin <i>HDAC 1/2</i>	ORR 25%, DOR 17 mo. PTCL indication withdrawn 2021 after Ro-CHOP was negative — still NCCN-listed and still FDA-approved for CTCL
Duvelisib <i>PI3K-δ/γ</i>	PRIMO (JCO 2026, n=123): ORR 48% , CR 33%, but PFS 3.4 mo; AITL ORR 62%, CR 51% , PFS 8.3 mo. Not FDA-approved for PTCL (phase 3 TERZO ongoing)
Valemetostat <i>EZH1/2</i>	VALENTINE-PTCL01: ORR 44% (52/119). Approved in Japan (ATLL 2022, PTCL June 2024); not US-approved
Golidocitinib <i>JAK1</i>	JACKPOT8 B: ORR 44% , CR 24% (94% Asian). Conditional approval in China (June 2024); US fast-track only
ALK inhibitors	Crizotinib FDA-approved Jan 2021 for age 1–21 (ORR 88%); adults off-label. Alectinib, brigatinib, lorlatinib also active
Others	Gemcitabine, bendamustine (ORR 50%, PFS 3.6 mo), lenalidomide (AITL), azacitidine in TET2-mutant TFH , cyclosporine for AITL immune phenomena

Response rates are all single-arm and PFS is short — use these to bridge a responder to **allo-HCT**, the only strategy with a durable plateau in R/R PTCL.

TOXICITY AND MONITORING

- **Brentuximab: peripheral neuropathy** — hold for grade 3, dose-reduce for grade 2. **PML** (boxed warning); pancreatitis. **Never with bleomycin** (pulmonary toxicity)
- **Pralatrexate: mucositis** is dose-limiting and prevented by B12/folate; check CBC weekly; reduce for CrCl <30
- **HDAC inhibitors**: thrombocytopenia, nausea, fatigue, **QTc** — ECG, keep K and Mg replete. **Belinostat: UGT1A1 genotype** guides the dose
- **Duvelisib boxed warnings**: fatal infections, **colitis/diarrhea**, severe skin reactions, pneumonitis; transaminitis. Mandatory PJP prophylaxis; CMV monitoring
- **Asparaginase**: hypersensitivity, **pancreatitis**, hepatotoxicity, **hypofibrinogenemia and thrombosis**, hyperglycemia. Check fibrinogen, lipase, LFTs, glucose each cycle
- **HD-MTX (SMILE)**: hydration + **urine alkalinization (pH ≥7)**, MTX levels, leucovorin rescue; **glucarpidase** if delayed clearance with AKI; stop PPIs, TMP-SMX, NSAIDs around the dose
- **Ifosfamide**: mesna; watch for encephalopathy (methylene blue) and Fanconi-type nephrotoxicity
- **ALK inhibitors**: crizotinib — visual disturbance, transaminitis, QTc, bradycardia
- Prophylaxis: **acyclovir + TMP-SMX** with HDAC, duvelisib, SMILE, alemtuzumab; **entecavir if HBcAb+**

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REGIMEN DOSING

Regimen	Dosing
BV-CHP <i>ECHELON-2</i>	Brentuximab 1.8 mg/kg (max 180 mg) + cyclophosphamide 750 mg/m ² + doxorubicin 50 mg/m ² d1, prednisone 100 mg d1–5 — q21d x6–8. No vincristine. Primary G-CSF
CHOP / CHOEP	CHOP as above + vincristine 1.4 mg/m ² (max 2) d1; CHOEP adds etoposide 100 mg/m² d1–3 — q21d x6 (NLG used CHOEP-14)
Brentuximab (R/R)	1.8 mg/kg IV q21d up to 16 cycles; 1.2 mg/kg if Child-Pugh A or neuropathy grade 2
Pralatrexate	30 mg/m² IV push weekly x6 q7wk. Folic acid 1–1.25 mg daily from 7–10 d before until 30 d after; B12 1 mg IM q8–10 wk. Leucovorin mouthwash for mucositis
Belinostat	1,000 mg/m² IV over 30 min d1–5 q21d; 750 mg/m² if UGT1A1*28 homozygous
Romidepsin	14 mg/m ² IV over 4 h d1, 8, 15 q28d — K ≥4, Mg ≥2 before each dose; QTc
Duvelisib <i>off-label</i>	75 mg PO BID x2 cycles, then 25 mg BID, q28d continuous. PJP + HSV/VZV prophylaxis
Gemcitabine	1,200 mg/m ² d1, 8, 15 q28d x3–6
DeVIC + RT <i>2/3 dose</i>	Dexamethasone 40 mg d1–3 + ifosfamide 1,000 mg/m ² d1–3 (+ mesna) + etoposide 67 mg/m ² d1–3 + carboplatin 200 mg/m ² d1 — q21d x3, concurrent RT 50 Gy
DDGP	Dexamethasone 15 mg/m ² d1–5 + cisplatin 20 mg/m ² d1–4 + gemcitabine 800 mg/m ² d1, 8 + pegaspargase 2,500 IU/m² d1 — q21d x6
Modified SMILE	MTX 2 g/m ² d1 (leucovorin from 24 h) · ifosfamide 1.5 g/m ² + mesna, dexamethasone 40 mg, etoposide 100 mg/m ² d2–4 · pegaspargase 2,000 U/m² d7 once MTX <0.04 μM — q28d x2–6. Urine alkalinization for HD-MTX
AZT + IFN <i>ATLL</i>	High-dose oral zidovudine + daily SC interferon-α, continued indefinitely while responding — no randomized dosing standard ; follow the treating center's protocol

DDGP doses are from the Chinese trial protocol (Li, Clin Cancer Res 2016). Interferon alfa-2b (Intron A) is off the US market — pegylated IFN is the practical substitute.

KEY TRIALS

Trial	Design	Result and meaning
ECHELON-2 <i>Lancet 2019</i>	Phase 3, double-blind, n=452, CD30 ≥10%, 70% sALCL	BV-CHP vs CHOP: mPFS 48.2 vs 20.8 mo , HR 0.71; OS HR 0.66 ; 5-yr OS 70% vs 61%. Benefit concentrated in sALCL
Ro-CHOP <i>JCO 2022</i>	Phase 3, n=421, untreated PTCL	Romidepsin + CHOP: PFS 12.0 vs 10.2 mo, HR 0.81 — negative ; more grade 3–4 cytopenias. Ended the PTCL indication
NLG-T-01 <i>JCO 2012</i>	Single-arm phase 2, n=166, ALK+ excluded	CHOEP-14 → ASCT: 72% reached transplant; 5-yr PFS 44% , OS 51%, TRM 4%
LYSA registry · COMPLETE	Observational, n=269 · n=119 in CR1	ASCT in CR1: propensity-matched no benefit (HR 1.02) vs OS HR 0.37 (driven by AITL). Unresolved
PROPEL <i>JCO 2011</i>	Single-arm, n=115	Pralatrexate: ORR 29%, DOR 10.1 mo, PFS 3.5 mo — accelerated approval, unconfirmed
BELIEF <i>JCO 2015</i>	Single-arm, n=129	Belinostat: ORR 26%, CR 11%, DOR 13.6 mo, PFS 1.6 mo
PRIMO <i>JCO 2026</i>	Single-arm, n=123	Duvelisib: ORR 48%, CR 33%; AITL ORR 62%, CR 51% . Not FDA-approved for PTCL
VALENTINE-PTCL01	Single-arm, n=133	Valemetostat: ORR 44%. Approved in Japan only
JACKPOT8 B	Single-arm, n=104, 94% Asian	Golidocitinib: ORR 44%, CR 24%. China only
JCOG0211 <i>JCO 2009</i>	Phase 1/2, n=33, stage IE–IIE nasal	DeVIC + RT 50 Gy: 2-yr OS 78% vs 45% historical RT alone
DDGP vs SMILE <i>JAMA Oncol 2022</i>	Randomized, n=87, stage III–IV NK/T	PFS HR 0.42, OS HR 0.41; 5-yr OS 74% vs 52%; less myelosuppression
JCOG9801 <i>JCO 2007</i>	Phase 3, n=118, aggressive ATLL	mLSG15 vs CHOP-14: CR 40% vs 25%; 3-yr OS 24% vs 13% (NS)
AZT/IFN meta-analysis <i>JCO 2010</i>	n=254, retrospective	First-line antivirals: 5-yr OS 46%; benefit in leukemic subtypes, not lymphoma type